



Diffusion model-based synthesis of brain images for data augmentation

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ABSTRACT

Training robust image segmentation models, particularly for medical applications like brain lesion segmentation, necessitates large, accurately annotated datasets. However, acquiring such datasets is often expensive and time-consuming. To address this challenge, we propose leveraging generative image models, specifically diffusion models, to synthesize realistic MRI brain images.

This study investigates the efficacy of two approaches: utilizing ControlNet and employing a custom-designed diffusion model. We evaluate the impact of these synthetic images on U-Net model training in two scenarios: (i) training on real, artificial, and mixed datasets of equal size, and (ii) incrementally augmenting a small real dataset with increasing amounts of synthetic data. We also benchmark the diffusion models against GAN-based competitors to provide a more comprehensive evaluation of their performance.

Our findings demonstrate that both diffusion models effectively generate high-quality brain images outperforming GANs in this task. Specifically, synthetic data generated by the diffusion models effectively approximates real brain image data, with minimal impact on segmentation performance, with only a 1.5% performance loss observed. Also, it significantly improves the U-Net's brain lesion segmentation performance when incorporated into the training process. Finally, our findings demonstrate that diffusion-generated images are more effective than GAN-generated images for improving segmentation model training.

This research highlights the potential of diffusion models to overcome data limitations in medical image analysis and pave the way for more accurate and efficient lesion segmentation.

1. Introduction

The field of medical image analysis is vital for advancing healthcare, with a wide range of applications from classifying breast cancer [1] to diagnosing Alzheimer's disease from Magnetic Resonance Imaging (MRI) scans [2]. Building on these efforts, our work focuses on brain lesion segmentation, a crucial task with significant implications for diagnosis, prognosis, and treatment planning [3–6]. The critical step of developing and refining artificial intelligence (AI) models for this task is significantly hindered by the limited availability of large-scale, well-annotated datasets. Acquiring a large dataset for brain lesion segmentation can take months or even years, as it requires recruiting patients and obtaining MRI or Computed Tomography (CT) scans. The need for expert radiologists to review and meticulously label images increases the time investment. Data augmentation techniques have been explored to address these limitations, with image generative models emerging as promising candidates [7–10].

While Generative Adversarial Networks (GANs) [11] have been widely employed for data augmentation in medical imaging, including image brain data synthesis, their limitations, such as mode collapse, instability and inability to capture intricate anatomical details, can hinder

their effectiveness. Diffusion models, on the other hand, have demonstrated superior performance in generating realistic, high-quality, diverse samples across various domains, including image synthesis [12–15]. However, their application to medical image synthesis, specifically artificial image augmentation for brain lesion segmentation is yet to be extensively investigated.

With the above mentioned in mind, this work aims to explore the potential of diffusion models for data augmentation in brain lesion segmentation. We present a novel method for brain lesion segmentation that leverages diffusion models to generate synthetic training data. Specifically, we explore two approaches utilizing image generative diffusion models. First, using ControlNet [16], we convert semantic segmentation maps of brain and lesion areas into realistic MRI brain scans. Next, we compare ControlNet's performance to that of our custom diffusion model trained for the same task. To provide a broader comparison, we also evaluate these diffusion-based generative approaches alongside popular GAN-based methods, namely SPADE [17] and Pix2Pix [18]. These synthetic MRI-segmentation pairs are then used to train a standard U-Net model for accurate brain lesion segmentation.

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In a broader context, our study contributes to the understanding of diffusion models' effectiveness in medical image augmentation. It also provides insights for future research in brain lesion segmentation. Our specific contributions are as follows:

- We are among the first to employ ControlNet to synthesize MRI brain images to augment training data.
- We propose a novel diffusion model designed for the same task.
- We compare diffusion-based generative models with GAN-based approaches, including SPADE and Pix2Pix.
- We evaluate the performance of the U-Net model trained on varying proportions of real and synthetic MRI brain images generated by the two diffusion models and two GAN models for brain lesion segmentation.
- We demonstrate synthetic data effectiveness in improving brain lesion segmentation accuracy, particularly when real data is limited.
- To ensure the reproducibility of our experiments, we publicly share the source code of our diffusion model, along with the scripts for all models training and evaluation used in this paper, in the GitHub repository: <https://github.com/Miretpl/synthetic-brain-mri-project>.

This paper is organized as follows. Section 2 reviews related work on medical image augmentation, including research on synthesizing brain images and diffusion models. Section 3 describes the brain image dataset used. Section 4 details the experimental pipeline, including diffusion models, GAN models and the segmentation model. Section 5 presents the results of image synthesis and its impact on segmentation performance. Section 6 discusses the findings, and Section 7 concludes the paper.

2. Related works

2.1. GAN-based methods

Generative Adversarial Networks have become a powerful tool for data augmentation in medical imaging. By synthesizing realistic medical images, GANs help overcome the challenges of limited labeled datasets and variability across imaging modalities. This has led to successful applications diagnosing in various organs, including the kidney, spleen [19], liver [20], lung [21], heart [22], and retinal [23] images. GAN-generated images have significantly improved image classification and analysis.

GANs have also been widely used to synthesize brain images. Early efforts mainly used unconditional GANs, such as vanilla GANs and their variants [24–28]. These models generate images from random noise, demonstrating the potential for data augmentation in brain image classification. For example, SAG-GAN uses semi-supervised attention-guided modules to synthesize plausible brain tumor images, improving ResNet18's accuracy more than traditional augmentation techniques [29]. However, these early models struggle to capture the intricate details of brain images [26], and lack the ability to generate images with specific characteristics, such as the presence of particular pathologies.

To overcome these limitations, conditional GANs, such as Pix2Pix [18], Pix2PixHD [30] and Spatially-Adaptive Normalization (SPADE) [17], have emerged for image-to-image translation. These models can learn complex mappings between paired images, enabling applications like generating different modalities from source modality [31], synthesizing brain images with realistic tumor representations from labeled inputs, and performing image segmentation by generating labels from input images [32]. Their versatility has led to widespread adoption in the medical imaging domain. For example, Pix2Pix-based approaches have been explored in studies such as [33, 34], while SPADE has been applied in works including [35–37]. However, the requirement for paired datasets, which are often expensive

and time-consuming to acquire, has limited their widespread application. CycleGAN [38] and its variants were developed, enabling unpaired image-to-image translation through cycle consistency constraints. This breakthrough expanded GAN applications to include cross-modality translation in multimodal MRI images [39,40], brain lesion segmentation [41–43], data augmentation [44,45], and the generation of synthetic data in scenarios where acquiring specific modalities is challenging [10]. Recent research has focused on incorporating anatomical priors and constraints into GANs to improve the realism and accuracy of synthesized brain images [46].

Despite these advancements, challenges remain. A comparative study across different GANs and medical imaging found that GANs still struggle to generate images with the same richness as real datasets, despite competitive Fréchet Inception Distance (FID) scores [47]. Additionally, GAN augmentation does not always improve the performance of Convolutional Neural Networks (CNNs) models for specific problems, like brain tumor image classification [28].

This highlights the need for continued research in more sophisticated generative architectures for specific medical imaging tasks.

2.2. Diffusion-based methods

Diffusion-based models [13,15] have shown remarkable success in various image-generation tasks, including medical imaging. These models generate high-quality, realistic images through an iterative denoising process, offering a potential solution to data scarcity, imbalance, and privacy concerns.

2.2.1. General applications

Diffusion models have been effectively used for dataset augmentation in general image-processing tasks. Beyond general image generation, diffusion models have demonstrated significant potential in controlled image synthesis. Controllable diffusion models, for instance, can generate diverse image variations while preserving original annotations, leading to improved object detection performance on benchmarks such as COCO and PASCAL VOC [48]. These models have also outperformed GANs in specific domains, such as weed classification, leading to significant enhancements in classification accuracy [49].

Image-to-image translation with diffusion models boosted few-shot classification accuracy by over 10% on augmented COCO datasets [50]. Moreover, Stable Diffusion-based augmentation has resulted in minor but consistent gains in image captioning tasks [51]. Beyond visual applications, diffusion models have been used to augment network data, enhancing protocol-constrained traffic generation and improving classification accuracy in network security [52].

2.2.2. Medical applications

In medicine, diffusion models address scarcity and improve diagnostic tools. For seizure prediction, diffusion models generated diverse Electroencephalogram (EEG) data, reducing imbalance and improving performance [53]. In oncology, feature augmentation using diffusion models showed promise in improving the performance of multiple instance learning algorithms for diagnosing prostate, colorectal, and breast cancer [54]. Custom diffusion models have been used to generate high-fidelity 3D MRI (breasts, knees, brain) and CT (chest) images, significantly improving the performance of downstream segmentation models [55]. Similarly, a Swin-transformer-based diffusion network created highly realistic synthetic 2D medical images, with classification results on synthetic datasets being comparable to those on real data [56].

Diffusion models have consistently outperformed traditional GANs in generating high-quality medical images. For lung ultrasound images, diffusion models have produced superior image quality compared to GANs [57]. Models like Medfusion have also outperformed GANs across a wide range of modalities and datasets [58]. Additionally, semantic 3D medical image synthesis with Med-DDPM improved upon

GANs, providing more stable, diverse, and anatomically coherent image generation [59].

Despite these promising advancements, the application of diffusion models for brain image augmentation remains relatively under-explored. This presents a significant opportunity for our research to use them for generating realistic and diverse brain images, which could lead to advancements in brain disease research and clinical practice.

2.3. ControlNet-based methods

ControlNet [16] is a flexible framework for conditioning generative diffusion models on various inputs such as segmentation maps, text prompts, and pose information. Its adaptability has led to significant advancements in different applications.

2.3.1. General applications

ControlNet has proven useful in creative and technical domains. For example, it enhanced text-to-music generation with time-varying controls, achieving more faithful melodies than MusicGen [60]. In art, it extracted stylistic features of JieHua paintings, achieving superior FID scores than CycleGAN [61].

Multimodal frameworks like C3Net integrated ControlNet to generate interconnected outputs like images, text, and audio, outperforming competitors in metrics such as FID and SPIDeR [62]. Material generation tasks have also benefited, with ControlNet consistently improving Root Mean Square Error (RMSE) and Structural Similarity Index Measure (SSIM) scores under variable lighting conditions [63]. While frameworks like OmniControlNet extended their capabilities to multi-condition inputs, their performance remained comparable to existing methods [64].

2.3.2. Medical applications

ControlNet has been instrumental in generating synthetic datasets and improving diagnostic tools in medical imaging. For instance, it facilitated the generation of pulmonary nodule X-ray images from segmentation maps with sufficient fidelity to train image recognition models [65]. Generated 3D CT images with the MAISI framework by incorporating ControlNet, outperforms baseline models in FID scores and enhances segmentation accuracy for tumors (liver, lung, pancreas, colon) and bone lesions by up to 7.7% when synthetic data was mixed with real datasets [66]. Similarly, it enabled cardiac MRI image generation using patient metadata and cardiac geometry, with models trained on mixed datasets achieving almost the same accuracy as those trained on real datasets [67].

ControlNet also supported pose-guided human movement image generation for rehabilitation, achieving moderate fidelity but improved Inception score (IS) scores [68]. In dermatoscopic imaging, it surpassed Pix2PixHD in metrics such as peak signal-to-noise ratio (PSNR) and SSIM in the generation, boosting lesion segmentation Dice-Sørensen coefficient (DICE) scores by around 5% [69]. Additionally, it excelled in laparoscopic image translation and text-guided segmentation tasks, improving metrics like DICE and Hausdorff distance-95 (HD95) on multiple datasets [70,71].

Importantly, to the best of our knowledge, ControlNet has not been explicitly applied to synthesize brain images for data augmentation in the medical imaging literature. This presents a unique opportunity to explore its potential in this domain.

3. Experimental data

This study used MRI brain images from the public BRA-TS2021 dataset [72–74]. The dataset comprises 3D MRI brain scans of 1251 patients with a $240 \times 240 \times 155$ resolution. Four scans acquired using different MRI methods (FLAIR, T1, T1CE, and T2-weighted) are available for each patient. Additionally, a lesion segmentation mask is provided for each patient. Each pixel in the segmentation map has

Table 1

Data subset characteristics for generative models.

Parameter	Subset		
	Train	Validation	Test
Num. patients	1000	126	125
Num. images	6000	756	750
Non-pathological samples (%)	46.70	47.49	46.27

assigned one of four labels, including no pathological change, necrotic tumor core, the peritumoral edematous tissue, and growing glioblastoma multiforme. We extended the masks with an additional label representing the brain region so that the generative models can see the complete brain outline, not only the lesions. The brain masks were obtained through thresholding and basic morphological operations on the original images.

In the experiments, FLAIR MRI images and their corresponding segmentation maps were selected from the available data. For each 3D scan, slices with a 20–140 index were chosen as they contain significant image information. These slices were sampled with a step of 20 to introduce diversity. As a result, six slices were extracted from each 3D source image. The selected slices were cropped to a resolution of 160×224 pixels, removing non-informative border pixels. This process yielded 7506 MRI image-segmentation mask pairs for 1251 patients. Among these, 3508 images (46.78%) contained no pathological changes.

A sample input image and the resulting labels are presented in Fig. 1.

Finally, the dataset was divided into training (80%), validation (10%), and testing (10%) sets. The split was performed at the patient level. Table 1 provides a quantitative summary of the datasets, including the percentage of images without pathological changes. These images were included as a negative control to identify potential false positives.

4. Methods

4.1. General pipeline

Our proposed approach utilizes diffusion-based generative models for data augmentation in brain lesion segmentation. The overall workflow is shown in Fig. 2. The pipeline begins with data preparation, where the real MRI dataset is split into training, validation, and test sets. Next, in the generative modeling stage, the training set is used to train diffusion models that learn to synthesize realistic MRI images conditioned on lesion segmentation maps. The resulting synthetic images are then combined with real data to form mixed training sets. Finally, in the segmentation modeling stage, U-Net-based segmentation models are trained on real, synthetic, and mixed datasets, and their performance is evaluated on held-out test data. This design directly compares segmentation performance across different training regimes and assesses the synthetic images' contribution to lesion segmentation.

To assess the effectiveness of the generative models as image augmentation technique, we train a U-Net model to perform brain and lesion segmentation using varying proportions of real and synthetic data. Specifically, we compare three training setups: (i) training the U-Net model exclusively with real data (*real* dataset), (ii) training with a 50/50 mix of synthetic and real data (*mixed* dataset), and (iii) training solely with synthetic data (*synthetic* dataset). Furthermore, we explore how increasing synthetic data to augment a small real image dataset affects U-Net image segmentation training. In all experiments, we compare the performance of our custom diffusion model with that of the established ControlNet architecture for image-to-image translation. We further compare both approaches with two GAN-based architectures, i.e., SPADE and Pix2Pix. For consistency, both networks were adapted to match the input-output data of the diffusion models, and the corresponding implementations are made publicly available in the accompanying GitHub repository.

We detail our experimental pipeline in the following sections.

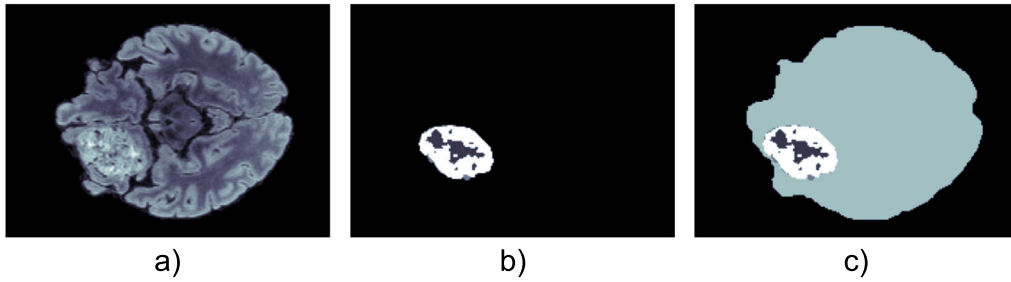


Fig. 1. Sample image and segmentation mask modification; (a) FLAIR image; (b) brain lesion segmentation map; (c) segmentation map with brain region and lesion.

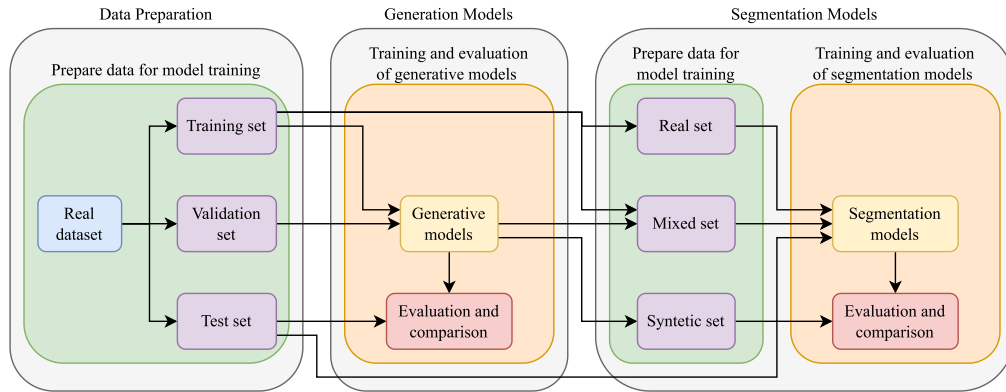


Fig. 2. General pipeline of work presented in this study.

4.2. Diffusion models

4.2.1. ControlNet

Model architecture. ControlNet is a generative model architecture built on Stable Diffusion, designed to incorporate additional conditioning parameters into the data generation process using diffusion networks [16]. As illustrated in Fig. 3, the network comprises an encoder, a z-space, and a decoder, highlighted in blue. The purple section represents the Stable Diffusion model, internally employing a U-Net architecture. A duplicate of the original network, excluding the decoder, is integrated into ControlNet via convolutional layers with a filter size of 1. Data from the z' space is processed through these convolutional layers and subsequently combined with data from the z space. Skip-connections within the U-Net model, originating from the encoder, are similarly treated by convolutional layers with a filter size of 1 and directed to corresponding decoder blocks. Data from these skip-connections, both from the original encoder and the modified encoder', are combined.

The text parameter, which serves as the conditional text for data generation by the Stable Diffusion model, is denoted by c_1 . The additional conditioning parameter is represented by c_2 , while x symbolizes the input data. The variable t indicates the step number within the diffusion process. The c_2 parameter is passed through a convolutional layer with a filter size of 1, and its processed data is then combined with the input data x before being fed into the modified encoder'. The output of the network is denoted by y .

During training, ControlNet learns to predict the noise added to the input data using the parameters t , c_1 , c_2 , and x . The model's error in noise prediction is calculated for the entire network, but weight adjustments are made exclusively to the duplicate of the original network. To enhance the model's ability to recognize distinctive features in the image conditioning c_2 , 50% of the c_1 values are randomly replaced with empty values during training.

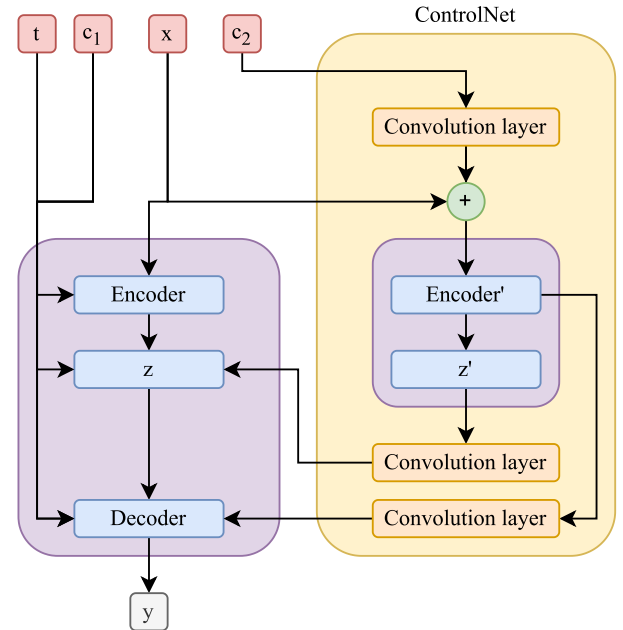


Fig. 3. The ControlNet architecture.

Training setup. The ControlNet training process was divided into three stages, i.e., (i) autoencoder training using Kullback–Leibler regularization, (ii) Stable Diffusion model training, and (iii) ControlNet training to integrate the additional conditioning parameters. Train and validation data (cf. Table 1) were used at this stage.

Each stage utilized 8 samples from the training set per series. The autoencoder, Stable Diffusion model, and ControlNet were trained for 100, 150, and 150 epochs, respectively. The model's performance was

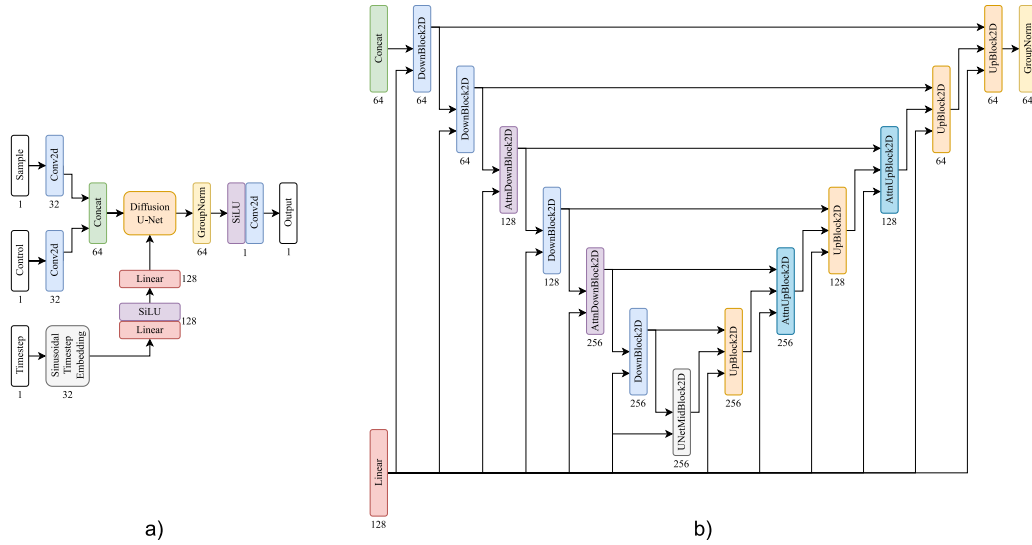


Fig. 4. The diffusion model proposed in this study; (a) general overview; (b) Diffusion U-Net details.

evaluated on a validation set every five epochs. If the loss function decreased, the model weights were saved.

Data reading was parallelized across 2, 4, and 4 processor cores for the autoencoder, Stable Diffusion model, and ControlNet. For the autoencoder, adversarial learning was initiated after the first 10 epochs (beginning of discriminator training). To ensure unit variance, the autoencoder results were multiplied by 0.3 during the training of the diffusion network and ControlNet [13].

The hyperparameters were refined through an iterative process, concentrating on the batch size, inference timesteps, and guidance scale.

4.2.2. Custom diffusion model

Model architecture. Our proposed diffusion model for synthesizing MRI brain images from segmentation maps is illustrated in Fig. 4. Fig. 4a presents a simplified overview of the model's architecture and Fig. 4b details the internal structure of the Diffusion U-Net block. The number of output channels for each layer or block is indicated below each segment.

The model's input consists of a brain scan and a segmentation map. Inputs are processed through convolutional layers. The outputs are concatenated and passed to the Diffusion U-Net model. Each convolutional layer utilizes 3×3 filters with a padding of 1. The resulting data, processed by the Diffusion U-Net block, is further refined using Group Normalization, nonlinear activation SiLU, and convolutional operations.

The Diffusion U-Net block begins with multiple DownBlocks2D in the contracting path that reduces the spatial resolution. Downsampling blocks with attention layers are interspersed between DownBlocks2D to improve the model's ability to capture long-range dependencies. The network center features a UNetMidBlock2D, bottleneck of diffusion U-Net network for processing features at the lowest resolution. Upsampling blocks (UpBlock2D) in the expanding path mirror the DownBlocks2D and progressively increase the spatial resolution. Similarly to downsampling blocks, between them are blocks with attention mechanisms. Notably, the last DownBlock2D layer does not downsample the output image, and the last UpBlock2D layer does not perform upsampling. The architecture uses skip connections to concatenate the corresponding Down and Up Blocks outputs, allowing the network to recover fine-grained details lost during downsampling. The final layers output a prediction, i.e., a synthesized brain MRI image, based on the processed features. The DownBlock2D, AttnDownBlock2D, UNetMidBlock2D, UpBlock2D, and AttnUpBlock2D layers utilize the following

parameters: normalization epsilon of $1e-05$, SiLU activation function, and a dropout probability of 0.3. Each layer contains 4, 4, 1, 5, and 5 Resnet2D blocks, respectively. All layers incorporating the cross-attention mechanism utilize 8 learned parameters to guide the model's attention to relevant regions within the input data.

Timestep embeddings are used, processed via linear, Sigmoid Linear Unit (SiLU) and linear layers to condition the network, providing information on the diffusion timestep. The diffusion process step, a time-based parameter, is processed by the Sinusoidal Timestep Embedding block, following the approach described in [75]. This block converts the step value into a suitable representation for the model.

Training setup. The model was trained for 200 epochs using train and validation data (cf. Table 1). Due to the small number of samples in each batch, a gradient accumulation function was used. Weight changes were made every 8 batches. After making the corrections, all weights were clipped to 1. Every five epochs, the model was evaluated on the validation set. The model weights were saved if the loss function, i.e., the mean squared error, was lower than before. Additionally, every five epochs, a few example images were generated. The diffusion probabilistic model (DDPM) noise was used as the denoising scheduler to transform the input image in the diffusion process. This object implements the method described in [76]. The number of steps in the diffusion process was defined as 2000. The AdamW algorithm was used as the optimizer. The learning rate was adjusted during training. Specifically, in the first 500 steps, it was increased linearly from 0 to $1e-4$. Then, the value was decreased linearly to 0 over the remaining training steps. The number of training steps equals the product of the data number in the training set and the number of epochs.

Hyperparameter optimization was performed by systematically tuning key parameters. We investigated the effects of batch size, learning rate, warm-up learning steps, number of inference timesteps, floating-point precision, and different loss functions, including Mean Squared Error (MSE), Mean Absolute Error (MAE), Structural Similarity Index (SSIM), and Cross-Entropy.

4.3. Image segmentation model

Model architecture. A U-Net architecture was employed for MRI brain image segmentation in this study. The detailed network structure is visualized in Fig. 5. To streamline the presentation, recurring layer patterns are consolidated into blocks. Each block comprises identical layer configurations with varying input and output channel numbers.

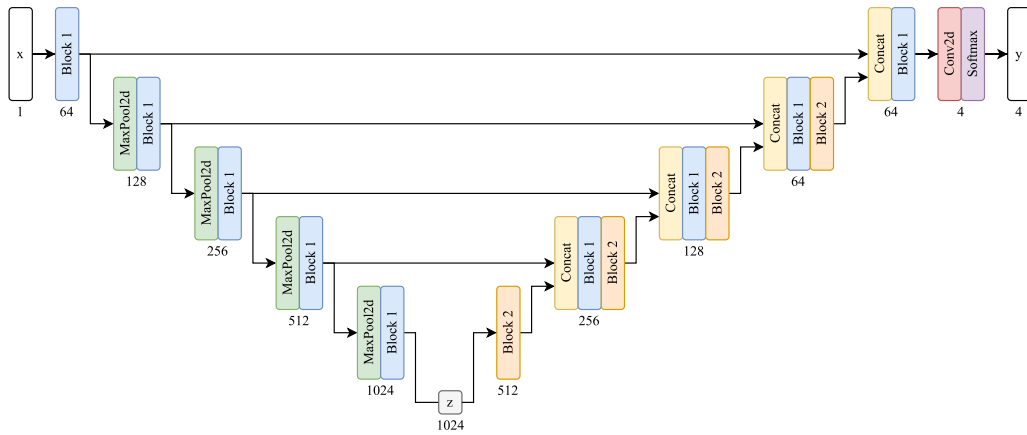


Fig. 5. The U-Net model used in this study for image segmentation.

Table 2

Datasets used for training the U-Net model with synthetic data for brain lesion segmentation.

Dataset	syn-50.0	syn-83.3	syn-90.0	syn-97.5	syn-98.7
Total num. images	20	60	100	400	800
Real images	10	10	10	10	10
[%] of synthetic images	50.00	83.33	90.00	97.50	98.75

The number of output channels is indicated beneath each network segment.

Block 1 sequentially stacks Conv2d, BatchNorm2d, ReLU, Conv2d, and ReLU layers. Convolutional layers utilize 3×3 filters with a padding of 1. Block 2 consists solely of a ConvTranspose2d layer employing 2×2 filters and a stride of 2, mirroring the configuration of the MaxPool2d layer. The final Conv2d layer, preceding the Softmax layer, has 1×1 filters.

Training setup. Two experiments were performed regarding the U-Net model. First, the U-Net model was trained on three distinct datasets: a *real dataset* composed solely of real images from the training set, a *mixed dataset* combining 50% real and 50% synthetic images, and a *synthetic dataset* consisting only of images generated by generative models. To evaluate the performance of our custom model, ControlNet and GAN-based networks in augmenting brain images for brain lesion segmentation, we created four variants of the *mixed* and *synthetic* datasets: one using images synthesized by our custom model, another using images generated with ControlNet, a third based on images produced by SPADE, and a fourth using images synthesized by the Pix2Pix model. In all cases, synthetic images were generated exclusively from brain masks selected from the training set. Each patient contributed three real and three generated images for the mixed datasets. The segmentation maps used to generate these latter images were chosen randomly.

In a second experiment, we trained a U-Net model using ten real images and a varying number of synthetic images (10 to 790), as detailed in Table 2. This experiment investigated whether augmenting training data with synthetic images improves U-Net performance. As in the first experiment, we used four sets of synthetic images for each training configuration: one generated by our custom model, second synthesized by ControlNet, another generated with SPADE and the last one produced by Pix2Pix. In all experiments, the test set was composed exclusively of real images.

Each time, the model was trained for 100 epochs. The Adam [77] was the optimizer. The learning rate was set to 0.0001. The focal loss [78] was used as the loss function.

4.4. Experimental setup

To train the ControlNet model, we utilized an implementation of the solution by [16] available in the GitHub repository¹. Our custom diffusion model and the U-Net segmentation model were developed in Python using PyTorch 2.0.1 and the Diffusers 0.25.0 library. For SPADE and Pix2Pix model, we adapted the configuration and source code provided in the official GitHub repositories^{2,3}.

The experiments were performed on a workstation equipped with an AMD Ryzen 7 5800X processor, 64 GB of HyperX Fury DDR4 memory operating at 3200MHz, an NVIDIA GeForce RTX 3080 graphics card with 12 GB VRAM, and a 1TB Samsung SSD 970 EVO Plus for storage.

5. Results

5.1. Image generation

Visual results. Sample visual results of brain scan synthesis using our custom diffusion model, ControlNet, SPADE and Pix2Pix are presented in Figs. 6–8. Specifically, Fig. 6 compares sample original brain images (f) to their synthetic counterparts generated by our custom diffusion model (b), the ControlNet (c), SPADE (d) and Pix2Pix (e) from the corresponding brain segmentation maps (a).

Fig. 7 illustrates the average of synthetic brain images and the MSE between the average real brain images and the averages of images synthesized by the custom model, ControlNet, SPADE and Pix2Pix. Darker regions indicate smaller differences in the corresponding parts of the average images, while lighter regions signify larger differences.

All models were verified in a diversity of image generation from a single brain mask. This diversity is illustrated in Fig. 8, which presents a sample brain image, its corresponding segmentation mask, and multiple brain images synthesized by our custom diffusion model (top panel), the ControlNet (second panel), the SPADE (third panel) and the Pix2Pix (bottom panel) from the mask.

Numerical results. The quality assessment of the synthetic images generated by all models was divided into two stages. The first stage involved assessing the similarity between the original images in the test set and the synthetic ones. Specifically, for each brain mask in the test set, one synthetic image was generated, and two metrics were used to evaluate their similarity: FID and Multiscale Structural Similarity Index Measure (MS-SSIM). FID compares the feature distribution of synthetic images with the feature distribution of real images, while MS-SSIM

¹ https://github.com/Warvito/generative_brain_controlnet

² <https://github.com/NVlabs/SPADE>

³ <https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix>

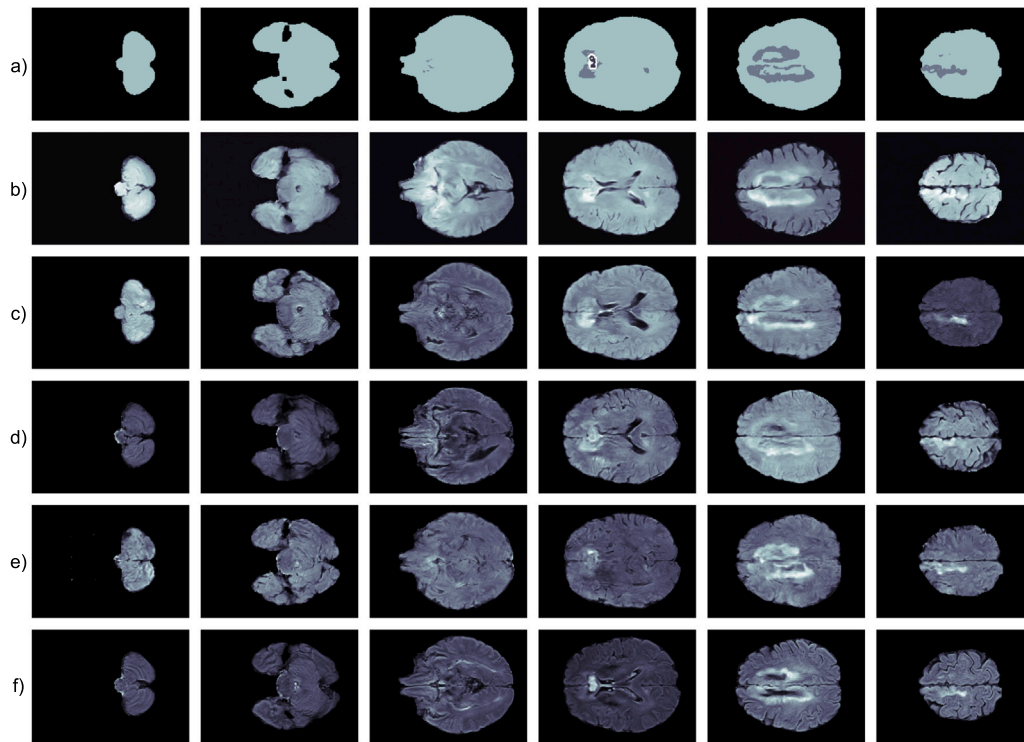


Fig. 6. Sample brain scans of one patient; (a) brain segmentation masks; brain images synthesized by: (b) the proposed diffusion model; (c) the ControlNet model; (d) the SPADE model; (e) the Pix2Pix model; (f) real brain images.

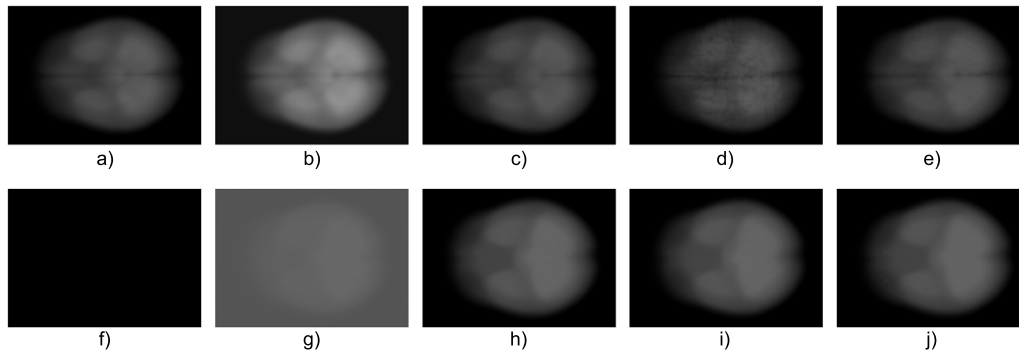


Fig. 7. Visual comparison of real and synthetic average images. (a) Average of real images; (b) Average of custom model-synthesized images; (c) Average of ControlNet-synthesized images; (d) Average of SPADE-synthesized images; (e) Average of Pix2Pix-synthesized images; (f) MSE between the real images and custom model-synthesized images; (g) MSE between the real images and ControlNet-synthesized images; (h) MSE between the real images and SPADE-synthesized images; (i) MSE between the real images and Pix2Pix-synthesized images; (j) MSE between the real images and real images.

pairwise compares the structural similarity of two images. FID has no predefined range. However, lower values indicate higher similarities between image distributions. MS-SSIM ranges from 0 to 1, with the latter indicating identical images. The diversity of images generated based on the same segmentation map was examined in the second stage. One thousand images were generated for one segmentation map, and again compared using the MS-SSIM metric. The results obtained in the experiment for all generative models are presented in [Table 3](#).

5.2. Image segmentation

Visual results. [Fig. 9](#) presents sample visual results of brain tumor segmentation. In this figure, we compare the ground truth results ([Fig. 9a](#)) with those obtained from the U-Net model trained with images

Table 3

Quantitative evaluation of synthetic images generated by the proposed, ControlNet, SPADE and Pix2Pix models using FID and MS-SSIM metrics.

Task	Metric	Model			
		Custom	ControlNet	SPADE	Pix2Pix
Reconstruction	FID	11.208	1.544	2.567	3.033
	MS-SSIM	0.682	0.771	0.781	0.779
Diversity	MS-SSIM	0.670	0.693	0.984	1.000

synthesized by four different methods: the purely synthetic data generated by our custom diffusion model ([Fig. 9b](#)), the ControlNet approach ([Fig. 9c](#)), the images generated by SPADE model ([Fig. 9d](#)) and the images generated by Pix2Pix model ([Fig. 9e](#)). Additionally, real brain

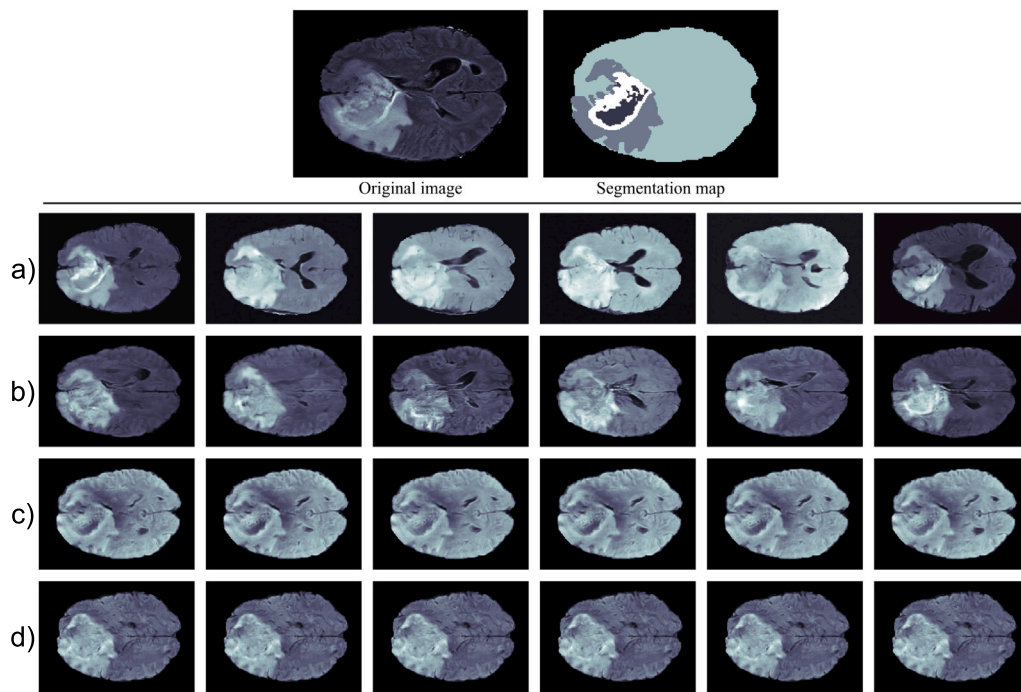


Fig. 8. Sample brain scans synthesized based on sample segmentation map; (a) synthesized by the proposed diffusion model; (b) synthesized by the ControlNet model; (c) synthesized by the SPADE model; (d) synthesized by the Pix2Pix model.

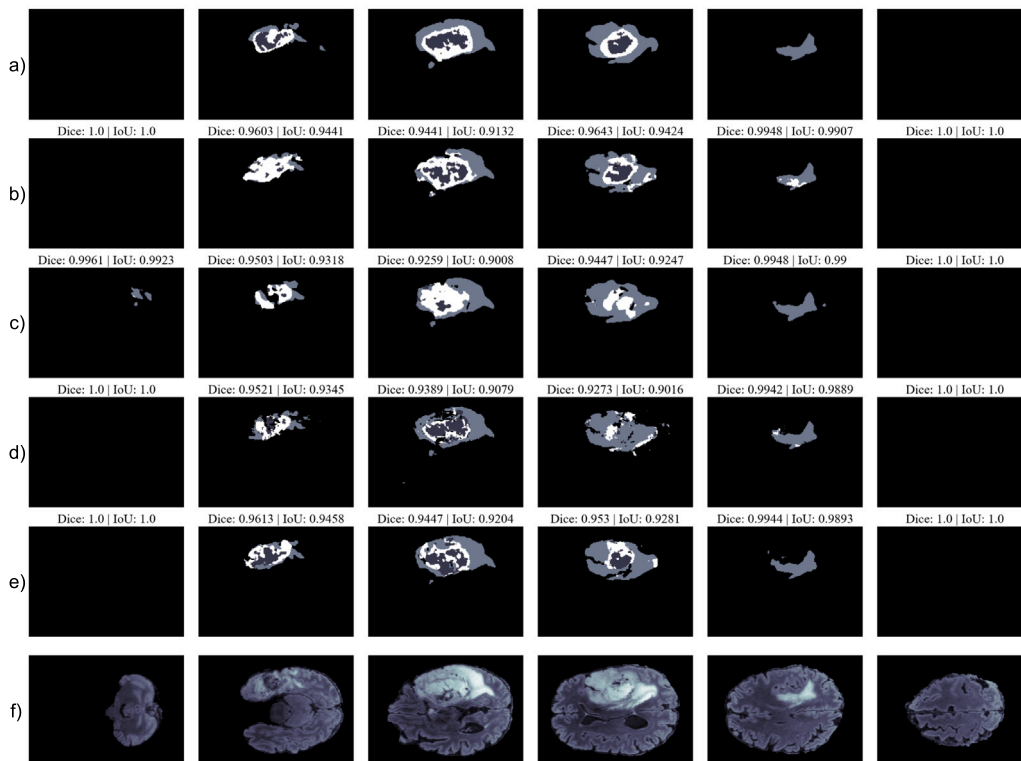


Fig. 9. Sample brain segmentation results of one patient; (a) ground truth segmentation results; (b) segmentation results obtained by the U-Net model trained with images synthesized by the proposed diffusion model; (c) segmentation results obtained by the U-Net model trained with images synthesized by the ControlNet; (d) segmentation results obtained by the U-Net model trained with images synthesized by the SPADE; (e) segmentation results obtained by the U-Net model trained with images synthesized by the Pix2Pix; (f) real brain images.

Table 4

Quantitative comparison of U-Net model performance on real, synthetic, and mixed (50/50) datasets obtained using custom, ControlNet, SPADE and Pix2Pix models, assessed using Intersection over Union (IoU) metric.

Score averaging	Patient group	Dataset								
		Real	Mixed				Synthetic			
			Custom	ControlNet	SPADE	Pix2Pix	Custom	ControlNet	SPADE	Pix2Pix
Micro	Healthy	0.9998	0.9999	0.9999	0.9998	0.9998	0.9999	0.9997	0.9889	0.9383
	Diseased	0.9692	0.9673	0.9646	0.9674	0.9669	0.9585	0.9574	0.9468	0.8888
Macro	Healthy	0.9765	0.9766	0.9809	0.9449	0.9651	0.9714	0.9729	0.6268	0.6542
	Diseased	0.5601	0.5399	0.5297	0.5330	0.5438	0.4679	0.4669	0.4298	0.4198
Weighted	Healthy	0.9999	1.0000	1.0000	0.9999	0.9999	0.9999	0.9998	0.9940	0.9643
	Diseased	0.9747	0.9732	0.9707	0.9730	0.9726	0.9669	0.9642	0.9598	0.9218

Table 5

Quantitative comparison of U-Net model performance on real, synthetic, and mixed (50/50) datasets obtained using custom, ControlNet, SPADE and Pix2Pix models, assessed using the DICE metric.

Score averaging	Patient group	Dataset								
		Real	Mixed				Synthetic			
			Custom	ControlNet	SPADE	Pix2Pix	Custom	ControlNet	SPADE	Pix2Pix
Micro	Healthy	0.9999	1.0000	1.0000	0.9999	0.9999	0.9999	0.9998	0.9940	0.9643
	Diseased	0.9842	0.9832	0.9817	0.9833	0.9829	0.9785	0.9779	0.9720	0.9352
Macro	Healthy	0.9766	0.9766	0.9809	0.9449	0.9651	0.9714	0.9729	0.6275	0.6584
	Diseased	0.6230	0.6028	0.5893	0.5959	0.6055	0.5302	0.5221	0.4908	0.4785
Weighted	Healthy	1.0000	1.0000	1.0000	0.9999	1.0000	1.0000	0.9999	0.9969	0.9806
	Diseased	0.9830	0.9819	0.9799	0.9819	0.9813	0.9771	0.9750	0.9732	0.9504

images are included for reference (Fig. 9f). For this presentation, we have selected what we subjectively consider to be the best, average, and worst cases. The corresponding DICE and IoU scores are provided above each segmentation result.

Numerical results. We compared the generated segmentation maps with ground truth annotations to evaluate the performance of the segmentation model trained on image data synthesized by the considered generative models. We employed two common metrics for this task: Intersection over Union (IoU) given by Eq. (1) and the DICE coefficient (cf. Eq. (2)).

$$IoU = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

$$DICE = \frac{2|A \cap B|}{|A| + |B|} \quad (2)$$

where A and B represent the image segmentation result and the ground truth, respectively.

To obtain a comprehensive performance assessment, we employed three averaging methods. Micro-average calculates the overall performance by averaging the metric scores across all image pixels. The macro-average evaluates the model's performance in each class individually. Weighted average accounts for class imbalance by weighting the class-wise scores based on their frequency. Furthermore, all results were averaged based on three separate segmentation model training rounds.

The IoU and DICE scores resulting from the first experiment are summarized in Tables 4 and 5, respectively. Each table compares the U-Net model's performance on real, synthetic, and mixed (50/50) datasets obtained using custom, ControlNet, SPADE and Pix2Pix models. The scores for diseased and healthy patients are calculated separately, to see if the U-Net model does not tend to find tumorous regions in healthy patients. The best scores are given in bold.

Figs. 10 and 11 illustrate the impact of increasing the proportion of synthetic data on U-Net performance for brain lesion segmentation. The plots display IoU and DICE scores from the second experiment,

where ten real images were supplemented with increasing amounts of synthetic data (50%, 83.3%, 90%, 97.5%, and 98.7%, labeled syn-50.0 to syn-98.7 as shown in Table 2). Results are shown for synthetic data generated by our proposed model (blue) and ControlNet (red), for both healthy (solid lines) and diseased (dashed lines) patients. For comparison, benchmark scores (green) are included, representing training with real images only, in quantities matching each synthetic dataset size (20, 60, 100, 400, and 800 images).

Statistical significance. We conducted pairwise comparisons across training regimes to evaluate whether synthetic data from different generative models affected segmentation performance. We first tested data distributions using the Shapiro–Wilk test for each evaluation metric (IoU and DICE in micro, macro, and weighted variants). None of the distributions met the normality assumption ($p < 0.05$), so all subsequent analyses were performed using the non-parametric Mann–Whitney U test. Comparisons were carried out between the baseline model trained only on real data and models trained with synthetic or mixed data from different generators and among the generators themselves. Specifically, the tests were conducted for models trained on purely synthetic datasets and mixed datasets (50% real + 50% synthetic), while expert data always refers to results obtained from models trained exclusively on expert-annotated real data. Statistical significance was determined at the level of $p < 0.05$. The results are summarized in Fig. 12 and 13 for synthetic and mixed data, respectively. In both figures, each subplot shows a heatmap where cells indicate whether the difference between a pair of methods is statistically significant (blue) or not significant (gray). The subplots are organized by evaluation metric and by healthiness condition, allowing direct comparison of method performance across different metrics and dataset types.

6. Discussion

Our results demonstrate diffusion models' potential for augmenting brain lesion segmentation data with the U-Net model, showing

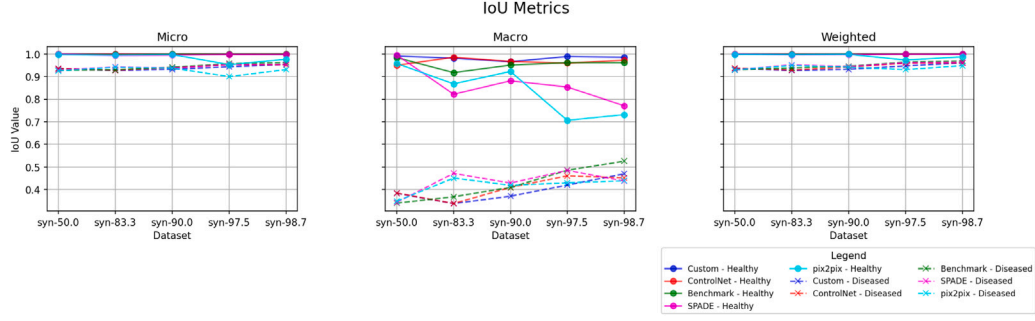


Fig. 10. IoU scores from the second experiment, showing U-Net performance for brain lesion segmentation with varying amounts of synthetic data. Results are shown for datasets containing 50%, 83.3%, 90%, 97.5%, and 98.7% synthetic data (syn-50.0 to syn-98.7) generated by our custom model (blue), ControlNet (red), SPADE (magenta) and Pix2Pix (cyan), for healthy (solid lines) and diseased (dashed lines) patients. Green lines represent benchmark scores obtained by training with real images only, using corresponding dataset sizes.

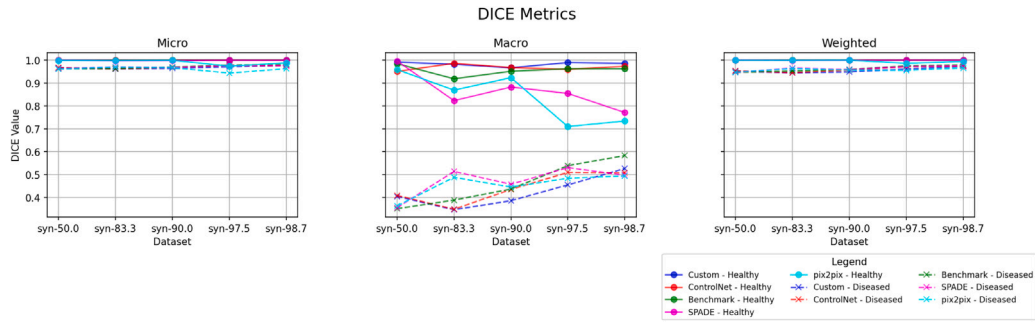


Fig. 11. DICE scores from the second experiment, showing U-Net performance for brain lesion segmentation with varying amounts of synthetic data. Results are shown for datasets containing 50%, 83.3%, 90%, 97.5%, and 98.7% synthetic data (syn-50.0 to syn-98.7) generated by our custom model (blue), ControlNet (red), SPADE (magenta) and Pix2Pix (cyan), for healthy (solid lines) and diseased (dashed lines) patients. Green lines represent benchmark scores obtained by training with real images only, using corresponding dataset sizes.

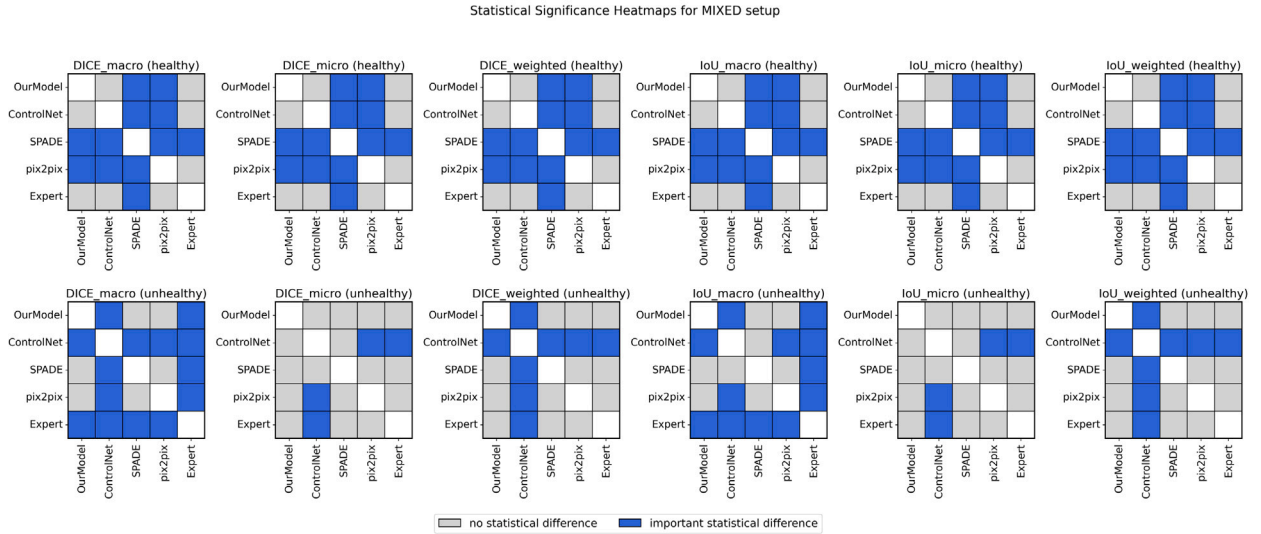


Fig. 12. Heatmaps showing pairwise statistical comparisons of segmentation performance across generative models on mixed data. Each subplot corresponds to a specific metric and healthiness condition. Blue cells indicate statistically significant differences between methods, while gray cells indicate no significant difference, based on Mann-Whitney U tests.

that diffusion-based augmentation consistently yields higher performance than GAN-based approaches, which tend to generate less diverse examples. The ControlNet and our custom diffusion model generate visually realistic and anatomically plausible brain images, as shown in Fig. 6 and 8. Despite only providing a rough brain outline, both models synthesize images with detailed anatomical features like the

ventricular system and cerebral cortex. Importantly, they accurately represent brain lesions, exhibiting visual characteristics consistent with real images. Similar observations can be drawn for GAN-generated images, although their anatomical fidelity is slightly lower. In particular, Pix2Pix tends to produce less consistent results, sometimes synthesizing brains with missing or simplified anatomical details (see Fig. 6e).

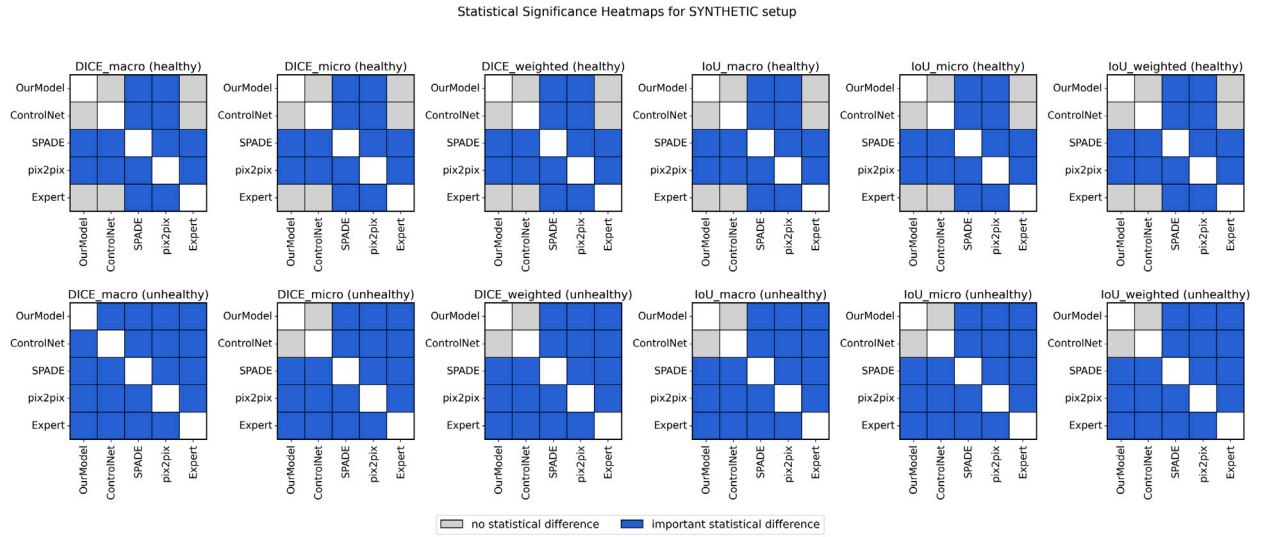


Fig. 13. Heatmaps showing pairwise statistical comparisons of segmentation performance across generative models on purely synthetic data. Each subplot corresponds to a specific metric and healthiness condition. Blue cells indicate statistically significant differences between methods, while gray cells indicate no significant difference, based on Mann–Whitney U tests.

Quantitative analysis supports these observations. MS-SSIM scores (cf. Table 3) indicate reconstruction abilities for all generative models, suggesting faithful preservation of brain structure.

Similarly, low FID scores confirm high similarity between real and synthetic images for all generative models. However, the custom model tends to produce brighter images than the other models, with a higher average intensity of 53.86 and standard deviation of 68.05 compared to the ControlNet’s average intensity of 28.65 and standard deviation of 45.26, SPADE’s average intensity of 26.00 and standard deviation of 41.59, and Pix2Pix’s average intensity of 28.13 and standard deviation of 42.27. At the same time the real dataset exhibits an average intensity of 29.63 and a standard deviation of 47.63. This discrepancy in intensity distribution also contributes to a higher FID score for the custom model. Additionally, the difference in FID score may be caused by the custom model *collapse* for some generations, which was not the case with ControlNet.

Visual inspection (cf. Fig. 7) reveals that the proposed model generates images with intensity distributions more similar to real brain images than ControlNet and both GAN-based models. This observation is supported by the lower MSE between the mean images of the real dataset and the custom-model synthesized dataset. Furthermore, the MSE visualization (cf. Fig. 7g) shows a more uniform error distribution across the custom model’s output, contrasting with ControlNet’s and GANs’ localized deviations, primarily concentrated in the brain region. This uniform error distribution may explain the proposed model’s superior performance, suggesting a focus on broader spatial relationships rather than individual pixel values. This, in turn, implies the potential learning of higher-level features and mitigation of image brightness variations, contributing to better generalization.

Both diffusion models exhibit satisfactory image variability when synthesizing from a single segmentation map (cf. Fig. 8). This diversity is confirmed by comparable MS-SSIM scores (cf. Table 3) for both models. It is crucial to note that FID may not be the most suitable metric for comparing medical images due to its training on ImageNet. Similarly, interpreting MS-SSIM requires careful consideration, as high values can indicate both high similarity and low variability. In contrast, GAN-generated images demonstrate substantially lower variability. For example, Pix2Pix can only generate a single image per input mask, as it lacks a mechanism for introducing diversity. This limitation is evident in Fig. 8c,d, where images produced by Pix2Pix and SPADE are nearly identical, which is further reflected in their MS-SSIM diversity scores

reaching 1 for both models (see Table 3). Consequently, GAN-based approaches offer limited capability for data augmentation compared to diffusion models.

Ultimately, the most reliable assessment lies in analyzing the impact of synthetic images on U-Net segmentation performance.

Our results demonstrate that training a U-Net segmentation model on mixed or purely synthetic data synthesized by diffusion models maintains comparable DICE and IoU scores to those achieved with real data alone (cf. Table 4, 5). For healthy patients, both IoU and DICE scores approach 1, indicating that all training scenarios effectively prevent false-positive tumor detections. Notably, using ControlNet-synthesized data slightly improves these scores compared to real data only. In contrast, GAN-based augmentations result in visibly lower performance across most variants, with Pix2Pix yielding the weakest outcomes, reflected in consistently reduced IoU and DICE values.

For diseased patients, the highest scores were obtained with real data. While synthetic data obtained from diffusion models slightly decreased scores and this effect was more pronounced with purely synthetic data, the disparities remain minimal. The difference in IoU (micro, weighted) does not exceed 1.5% from the best-performing scenario (real data only), and the DICE score difference remains below 1%. These findings strongly suggest that synthetic data generated by the diffusion model can effectively approximate real brain image data. At the same time, performance with GAN-based data is visibly worse. In particular, Pix2Pix exhibits the weakest results, with IoU (micro, weighted) scores lower by up to 6% for healthy patients and up to 8% for diseased patients compared to real-data training, while SPADE shows a similar but slightly less pronounced decline. For both GAN-based models, macro-level IoU and DICE scores drop significantly, even for healthy patients, where both diffusion models achieve the best performance. This contrast underscores the limitations of GAN-based augmentation, which fails to preserve consistent segmentation quality across patient groups and evaluation metrics. In contrast, diffusion-based models maintain stable performance but also generalize more effectively, making them a more reliable choice for medical image augmentation.

When diffusion models are considered, our custom model-synthesized data yields slightly better performance, with up to 0.3% higher IoU and 1.6% higher DICE scores compared to ControlNet-synthesized data. The observed difference in macro IoU and DICE suggests that the U-Net model trained on every dataset (real, mixed,

synthetic) performs worse on specific classes, likely due to class imbalance, particularly for classes represented by a few small objects. The necrotic tumor core is the least represented class in the dataset, appearing in approximately 27% of the images. The limited representation of this class poses challenges for the segmentation model, hindering its ability to learn effectively. To address this issue and enhance model performance, the loss function should be adapted to account for class imbalance, ensuring better learning and representation of underrepresented classes.

Figs. 10 and 11 demonstrate that U-Net-based brain lesion segmentation significantly benefits from the inclusion of additional training data, regardless of the model used. Notably, this enhancement occurs regardless of whether the data is real or synthetic. IoU and DICE scores consistently increase across all evaluated variants as training samples grow.

This trend is particularly pronounced in the case of diseased patients. When training with 10 real images augmented by 790 synthetic images, the macro DICE score increased by over 19% using ControlNet-synthesized brain images and by 23% using our custom model for data augmentation. The corresponding gains in macro IoU were approximately 15% and 18%, respectively. Micro and weighted IoU and DICE scores also benefited by approximately 2% from augmenting the dataset with synthetic data. While the advantage of augmenting small training datasets with ControlNet-generated data is slightly less pronounced than with our custom model, both diffusion models effectively enhance the performance of U-Net-based brain lesion segmentation from MRI images when applied to limited training data.

For diseased cases, all models, including GANs, contribute to improved segmentation performance. However, for healthy patients, GAN-based augmentation reduces accuracy in the macro IoU and macro DICE scores, where both metrics decrease compared to training on real data alone. This degradation may stem from the limited anatomical variability and lower fidelity of GAN-generated images, which can introduce unrealistic patterns or bias the model toward spurious features. In contrast, diffusion-based models appear to better capture the diversity and fine-grained anatomical structure of healthy brains, allowing them to provide synthetic samples that reinforce rather than weaken the segmentation model.

The statistical significance analyses (see Fig. 12 and 13) confirm the performance trends observed in raw IoU and DICE scores. In the case of mixed datasets, significant differences between methods are relatively rare, suggesting that the presence of real data buffers the impact of less effective generators. By contrast, statistically significant differences are widespread for purely synthetic datasets, particularly between diffusion-based and GAN-based approaches. This pattern demonstrates that GANs do not provide a reliable substitute for real data, while diffusion models approximate real-data performance closely. At the same time, comparisons between our custom diffusion model and ControlNet show few significant differences, indicating comparable effectiveness. Overall, the statistical analysis shows that the observed superiority of diffusion models is robust, systematic, and not attributable to random variation, further strengthening the case for their use in medical data augmentation.

7. Conclusions

This study demonstrates the potential of diffusion models, specifically the custom diffusion model and ControlNet, for augmenting medical datasets in the context of brain lesion segmentation. By synthesizing high-quality, anatomically realistic MRI brain images, these models effectively mitigate the challenges associated with limited data in medical image analysis.

Our findings show that synthetic data generated by diffusion models approximates the performance of models trained on real data, with a minor segmentation performance loss. Moreover, incorporating

synthetic images into the training process significantly enhances the accuracy of U-Net-based brain lesion segmentation, particularly when real data is limited. Between the two evaluated diffusion models, the custom diffusion model slightly outperformed ControlNet in generating images closer to the real data distribution, leading to better segmentation performance. In direct comparison, both diffusion models consistently outperformed GAN-based approaches, which produced less diverse and less anatomically coherent images, resulting in lower segmentation scores.

The results emphasize the robustness of diffusion models in generating diverse and realistic medical images while also providing a scalable solution for augmenting small datasets. This research paves the way for more accessible and efficient training of segmentation models, reducing reliance on extensive, annotated datasets.

Future work could focus on refining the synthetic image generation process to address specific challenges identified in this study. For instance, improving the intensity consistency of the proposed diffusion model could enhance its fidelity to real data and further reduce segmentation performance gaps. Additionally, optimizing the models to handle class imbalances, such as those associated with smaller lesion regions, could improve segmentation accuracy across all classes.

Expanding the scope of experiments to include other MRI modalities or multi-modality data, such as combining T1 and FLAIR sequences, could provide a broader validation of these models. Investigating alternative integration strategies for synthetic data, such as incremental augmentation schedules or adaptive mixing ratios, may reveal ways to maximize their utility in training workflows.

By addressing these areas, the methods presented in this paper could become even more effective tools for overcoming data limitations in medical imaging and advancing segmentation model performance.

CRedit authorship contribution statement

Przemysław Mirowski: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anna Fabijańska:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used ChatGPT and Grammarly in order to improve the language of some parts of the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Public image dataset was used in this study.

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